

A Comparison of Tree Based Machine Learning Algorithms in Modeling Cyclist Injury Severity



1 Introduction

Cyclists have always been a vulnerable group in vehicle motor crashes. According to the Fatality Analysis Reporting System (FARS)¹, from 2017-2019, total fatality rate is within 43% – 44%. Pedestrian’s fatality rate is within 93% – 94%. However, cyclist’s fatality rate remains as high as 96% – 97%. Therefore, understanding vehicle crashes involving cyclists is important. Common approaches in analyzing crash data and injury severity are statistical methods such as ordered probit models [1, 2, 3], multinomial logit model [4, 5, 6], and binary logit model [6]. Statistical methods are easy to interpret and have a clear formulation between dependent variables and independent variables. However, they also impose strong assumptions on the data [7].

With the development of machine learning methods, the researchers in transportation and civil engineering have steered their focuses towards applications of novel cutting-edge machine learning methods in injury analysis. Among different machine learning algorithms, tree-based methods have attracted a lot of attentions because their flexibility makes the predictive performance outstanding in real-world data analysis. For this project, I apply four different tree-based algorithms, random forest, multiclass AdaBoost, gradient boosting machine and XGBoost, on Crash Report Sampling System (CRSS) in order to understand their predictive performance in cyclists’ injury severity analysis. I also use in-built feature importance functions to understand contributions of risk factors in determining final injury levels. This project is implemented in Python.

¹<https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>

2 Data

This section first describes the data source that is used and the necessary process to construct a supervised learning problem.

2.1 Data Source

For this project, Crash Report Sampling System (CRSS)² is analyzed to understand cyclist injury severity in motor vehicle crashes. CRSS is public data made available by National Highway Traffic Safety Administration(NHTSA). It uses a multistage selection process to collect a nationally representative probability sample of police-reported crashes involving motor vehicles. Every crash reported in the CRSS must involve at least one motor vehicle and cause property damage, injury or death. It also provides comprehensive information such as vehicle information, information of drivers and others involved in the crash, and external conditions such as weather, light conditions, and road conditions. To be more specific, it consists of 20 data files and each file can be combined by using key columns. CRSS is reported annually, and three-year data, 2017-2019, is used.

2.2 Feature Selection, Feature Engineering and Problem Construction

Dependent Variable. As mentioned above, CRSS provides various information regarding the crash. Also, it provides person type and person injury severity level for everyone involved in the crash. Therefore, cyclists' injury severity is used as dependent variable Y , and corresponding information can be included in the feature space. The original dataset separated injury severity levels into eight categories: No Apparent Injury, Possible Injury, Suspected Minor Injury, Suspected Serious Injury, Fatal Injury, Injured, Severity Unknown, Died Prior Crash, Unknown/Not Reported. Among these categories, four categories are dropped. First, these classes have no or less than five records. Therefore, they can be easily seen as noises by the models. Second, they cannot combined with other classes. These four injury levels are Injured, Severity Unknown, No Apparent Injury, Died Prior Crash and, Unknown/Not Reported. For the rest of the classes, "Possible Injury" and "Suspected Minor Injury" are combined as "Possible/Minor Injury" because they indicate a similar level of injury severity. To conclude, the dependent variable Y has classes: Possible/Minor Injury,

²<https://www.nhtsa.gov/crash-data-systems/crash-report-sampling-system>

Suspected Serious Injury, and Fatal Injury. They contain 4361, 555, and 80 records, respectively.

Independent Variable. The features are first selected if they are deemed interesting to investigate. Then Goodman-Kruskal gamma[8] is used to eliminate highly correlated variables. Then Goodman-Kruskal gamma tests rank relationship between two ordinal variables by using concordant and discordant pairs. Even though they are labelled as categorical variable, the majority of features I choose contain implicit ordinal relationship. For example, light condition variable have categories: Daylight, Dark-lighted, and Dark-Not Lighted. It is clear that there is an implicit ranking indicting the light intensity. Similar arguments can be made for weather condition, vehicle body types, etc. However, there is no formal definition of gamma value thresholds, i.e., which value is considered high and which value is considered low. Therefore, for this project, I follow the rule of thumb stated in [9] for correlation coefficients, which is described in Table 1. The features which correlations are considered *Very Strong* are eliminated.

Table 1: Interpretation for Gamma Values

$ \gamma $ Range	Interpretation
$[0, 0.1)$	Negligible
$[0.1, 0.2)$	Weak
$[0.2, 0.4)$	Moderate
$[0.4, 0.6)$	Relatively Strong
$[0.6, 0.8)$	Strong
$[0.8, 1]$	Very Strong

After feature selection and feature engineering process, Table 2 describes the variables that are used for supervised learning.

2.3 Data Cleaning

Missing data. CRSS contains missing entries since it is based on police report. If a record is eliminated when it contains NA/Unknown entry, 93.2% data are dropped. Therefore, data imputation is needed. For this project, I use mode imputation.

Duplicate data. It is possible that one crash involves multiple vehicles and cyclists. If every record relating to this crash is included, information such as weather and road conditions will be double-counted. In order to present over-representing problem, only the first record of cyclists and

Table 2: Description of Variables

Variable Types	Variable Name	Variable Types	Description
Dependent Variable	INJ_SEV	Categorical	Injury Severity: 1=Possible/Minor Injury 2=Suspected Serious Injury 3=Fatal Injury
External Environment	LGT_COND	Categorical	Light condition at the time of the crash
	WEATHER	Categorical	Weather condition at the time of the crash
Crash Time	MONTH	Categorical	Month of Crash
	HOUR	Categorical	Hour of Crash
	DAY_WEEK	Categorical	The Day of the Week on which the Crash Occurred
Crash Location and Its Related Information	RELJCT2	Categorical	Crashes' Relation to Junction
	WRK_ZONE	Categorical	If the crash qualifies as a "Work Zone Accident" then the type of work activity is identified
	PBCWALK	Categorical	If a marked crosswalk was present at the crash site
	PBSWALK	Categorical	If a sidewalk was present at the crash site
	PBSZONE	Categorical	If the crash occurred in a school zone
	VSPD_LIM	Ordinal	Speed Limit
	VPROFILE	Categorical	Roadway Grade
	VTCONT_F	Categorical	Traffic Control Device Functioning
Driver and Vehicle Information	SCH_BUS	Categorical	If a school bus, or motor vehicle functioning as a school bus, is related to the crash
	BODY_TYP	Categorical	A classification of this vehicle based on its general body configuration, size, shape, doors, etc.
	P_CRASH1	Categorical	Pre-Event Movement (Prior to Recognition of Critical Event)
	MDRDSTRD	Categorical	Driver Distracted By
	DRIMPAIR	Categorical	Condition (Impairment) at Time of Crash- Driver
Cyclist Information	AGE	Ordinal	The age of the cyclist
	SEX	Categorical	The sex of the cyclist

vehicles is selected for every crash.

Dealing with categorical features. Since several model modules cannot process categorical features directly, all the categorical features are one-hot-encoded. The first column is not dropped because one of the project's purpose is to understand each feature/category's contribution to the final output, eliminating one column makes feature importance ranking harder to interpret.

3 Methodology

3.1 Machine Learning Algorithms

Machine learning models are extremely popular in recent years because of their strong predictive and computational power. Among various ML methods, tree-based models have received a lot of attention. Tree-based models do not impose strong assumptions on the data, which makes them good candidate for real-world data analysis. Meanwhile, they are easier to understand. For this project, four different tree-based models are used to fit in CRSS data, and *Scikit-learn* and XGBoost python packages are used for modelling. I also use grid search and 5-fold cross validation for hyperparameter tuning.

Random Forest. Random forest is an improvement over bagged trees. It randomly bootstrap sample data and build individual decorrelated trees. Compared to one single decision tree, random forest is not subject to overfitting problem even the number of built tree increases. Meanwhile, compared to bagged trees, it further decorrelates trees by randomly sample features that built trees. In the end, random forest makes decisions based on majority votes.

AdaBoost, Gradient Boosting Machine, and XGBoost AdaBoost, Gradient Boosting Machine, and XGBoost are based on boosting technique. Unlike bagging technique, which builds classifiers separately, boosting learns classifiers sequentially, and each classifier tries to correct mistakes made by previous one. AdaBoost builds weak learners. Each tree only has one node and two leaves. In a forest of weak learners learned by AdaBoost, some weak learners have more say than others in the final classification. Unlike random forest, AdaBoost is subject to overfitting problem if the number of learned trees increases. Difference between Binary AdaBoost and Multiclass AdaBoost is not trivial. For this project, I use multiclass AdaBoost from *scikit-learn*, which implements the algorithm based on [10]. Unlike AdaBoost, Gradient Boosting Machine and XGBoost normally learn full trees with fixed size and each tree has the same learning rate. Gradient Boosting Machine learns decision while XGBoost uses a unique classification tree. Therefore, in addition to hyperparameters relating to decision trees, XGBoost also has tunable parameters that regularize trees.

3.2 Evaluation Metrics

This project is a supervised learning multiclassification problem with imbalance dataset. There are three dimensions that I use to understand model performance. First, I use prediction accuracy. Accuracy is a common way to understand a model’s predictive power. However, it is not a good measure when we have an imbalance dataset. Therefore, I will also report macro and weighted average of precision, recall and F-1 scores. Second, I use training runtime for each model. Third, I report top 6 features that contribute the most. All four tree methods described in the previous section have in-built feature importance attribute. For random forest, AdaBoost, and Gradient Boosting Machine, the feature importance is based on impurity. For XGBoost, the standard I choose is *total gain*, which sums up the contribution of the corresponding features across all the trees during classification.

4 Results

The prediction and computation performance of each model are presented in Table 3. As we can see, XGBoost is fastest in term of training time, while gradient boosting machine outperforms in terms of all prediction performance metrics.

Table 3: Model Results Comparison

Model	runtime (s)	Accuracy	Macro			Weighted		
			Precision	Recall	F1-score	Precision	Recall	F1-score
RF	0.62	0.864	0.288	0.333	0.309	0.748	0.864	0.802
AdaBoost	1.40	0.865	0.455	0.349	0.338	0.759	0.865	0.804
GBM	1.37	0.868	0.703	0.402	0.416	0.841	0.868	0.814
XGBoost	0.25	0.865	0.455	0.349	0.338	0.759	0.865	0.804

The top 6 features based on feature importance ranking are presented in Table 4. As we can see, the results are similar and meet our expectation. The features that are selected by all four models are *Speed Limit*, *Age of the Cyclist*, and *Vehicle’s Precrash Movement: Going Straight*. This results makes sense because speed limit feature is an indicator of travel speed of the vehicle when crash happens. The vehicle is harder to stop when it is moving fast, and vehicle is more likely to stay high speed when it is going straight or negotiating a curve. Older group and young children are harder

to notice coming vehicles or react quickly. Similar arguments can also be made for other selected features. Big, heavy vehicles like *Single-Unit Truck* are hard to control and the drivers are easy to have blind spots because of their heights. Drivers under influence are unable to react quickly.

Table 4: Top 6 Most Important Features Selected by Different Tree-Based Model

Rank	Random Forest	AdaBoost	GBM	XGBoost
1	Age of the Cyclist	Speed Limit	Age of the Cyclist	Speed Limit
2	Speed Limit	Age of the Cyclist	Speed limit	Vehicle's Precrash Movement: Going Straight
3	Vehicle's Precrash Movement Going Straight	Vehicle's Precrash Movement Going Straight	Vehicle's Precrash Movement Going Straight	Relation to Junction Non-Junction
4	Vehicle's Body Type 4-Door Sedan	Vehicle's Body Type Single-Unit Straight Truck or Cab-Chassis	Relation to Junction: Non-junction	Age of Cyclist
5	Driver Distracted By: Not Distracted	Driver Distracted By: Looked but Did Not See	Vehicle's Precrash Movement: Negotiating a Curve	Vehicle's Precrash Movement: Negotiating a Curve
6	Driver Distracted By: Unknown	Time of Crash: 20:00-20:59	Driver Under the Influence of Alcohol, Drugs or Medication	Light Condition Daylight
7	Relation to Junction: Non-Junction	Driver Under Influence of Alcohol, Drugs and Medication	Light Condition: Daylight	Roadway Grade: Non-Trafficway or Driveway access

5 Discussion

Overall, the output meets the expectation. One thing that is out of expectation is performance of XGBoost is not as good as Gradient Boosting Machine. XGBoost is a newly developed implementation based on gradient boosting framework and has been high praised by the data science community. It tells us that we should always look at machine learning algorithms critically. Second, one problem with ensemble methods like bagging and boosting is they are black-box and it is hard to make meaningful interpretation from the results. For example, in this project, we only know the important features but we do not know how they affect the final results. For application problem like injury analysis, interpretation is always important.

References

- [1] Chris Lee and Mohamed Abdel-Aty. “Comprehensive analysis of vehicle–pedestrian crashes at intersections in Florida”. In: *Accident Analysis & Prevention* 37.4 (2005), pp. 775–786.
- [2] Naved A Siddiqui, Xuehao Chu, and Martin Guttenplan. “Crossing locations, light conditions, and pedestrian injury severity”. In: *Transportation research record* 1982.1 (2006), pp. 141–149.
- [3] Kelvin KW Yau, Hing-Po Lo, and Sherrice HH Fung. “Multiple-vehicle traffic accidents in Hong Kong”. In: *Accident Analysis & Prevention* 38.6 (2006), pp. 1157–1161.
- [4] Venkataraman Shankar and Fred Mannering. “An exploratory multinomial logit analysis of single-vehicle motorcycle accident severity”. In: *Journal of safety research* 27.3 (1996), pp. 183–194.
- [5] Ahmad Khorashadi et al. “Differences in rural and urban driver-injury severities in accidents involving large-trucks: an exploratory analysis”. In: *Accident Analysis & Prevention* 37.5 (2005), pp. 910–921.
- [6] Peter Savolainen and Fred Mannering. “Probabilistic models of motorcyclists’ injury severities in single-and multi-vehicle crashes”. In: *Accident Analysis & Prevention* 39.5 (2007), pp. 955–963.
- [7] Jian Zhang et al. “Comparing Prediction Performance for Crash Injury Severity Among Various Machine Learning and Statistical Methods”. In: *IEEE Access* 6 (2018), pp. 60079–60087. DOI: 10.1109/ACCESS.2018.2874979.
- [8] Leo A. Goodman and William H. Kruskal. “Measures of Association for Cross Classifications”. In: *Journal of the American Statistical Association* 49.268 (1954), pp. 732–764. ISSN: 01621459. URL: <http://www.jstor.org/stable/2281536>.
- [9] Louis M Rea and Richard A Parker. *Designing and conducting survey research: A comprehensive guide*. John Wiley & Sons, 2014.
- [10] Trevor Hastie et al. “Multi-class adaboost”. In: *Statistics and its Interface* 2.3 (2009), pp. 349–360.